**Unit 2 (Topic 3): All Definitions**

| Term | Definition |
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| Cell theory | Concept stating all living organisms are made of cells, cells are the basic units of life, and cells arise from pre-existing cells. |
| Eukaryotic cell | A cell with a membrane-bound nucleus and organelles (e.g. animal, plant cells). |
| Prokaryotic cell | A cell without a nucleus or membrane-bound organelles (e.g. bacteria). |
| Ultrastructure | Detailed internal structure of a cell visible only under an electron microscope. |
| Organelles | Structures within a cell specialised for specific functions (e.g. mitochondria, ribosomes, Golgi). |
| Magnification | The degree to which an image is larger than the actual object. |
| Resolution | The ability to distinguish two points as separate; clarity of an image. |
| Mitosis | Nuclear division producing two genetically identical daughter cells. |
| Meiosis | Type of nuclear division producing four genetically unique haploid gametes. |
| Cell cycle | Series of stages (interphase and mitosis) through which a cell passes from one division to the next. |
| Interphase | Cell cycle phase (G1, S, G2) where the cell grows, replicates DNA, and prepares for division. |
| Gamete | Haploid sex cells (sperm and egg) formed by meiosis, used in sexual reproduction. |
| Fertilisation | Fusion of haploid gametes (egg and sperm) producing a diploid zygote. |
| Zygote | Diploid cell formed by fertilisation, develops into an embryo. |
| Stem cell | Undifferentiated cell capable of differentiating into various cell types. |
| Totipotent stem cell | Stem cell able to differentiate into all cell types, including placental tissue (e.g. zygote, early embryonic cells). |
| Pluripotent stem cell | Stem cell able to differentiate into almost all cell types (excluding placental tissue, e.g. embryonic stem cells). |
| Morula | Early embryonic stage consisting of a solid ball of undifferentiated totipotent cells. |
| Blastocyst | Hollow ball of cells containing a pluripotent inner cell mass. |
| Differentiation | Process by which unspecialised cells become specialised for specific functions. |

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| Differential gene expression | Expression of specific genes leading to cell specialisation. |
| Epigenetics | Study of heritable changes in gene expression without alteration of DNA sequence (e.g. methylation, histone modification). |
| DNA methylation | Addition of methyl groups to DNA, typically suppressing gene expression. |
| Histone modification | Chemical changes to histone proteins affecting gene expression by altering DNA packaging. |
| Polygenic inheritance | Traits controlled by multiple genes, resulting in continuous variation (e.g. height, skin colour). |
| Continuous variation | Variation in a population showing a range of phenotypes, not discrete categories. |
| Phenotype | Observable physical and biochemical characteristics of an organism determined by genotype and environment. |
| Genotype | Genetic makeup of an organism; combination of alleles. |
| Mutations | Random changes in DNA sequence, potentially altering gene expression and phenotype. |
| Substitution mutation | Mutation replacing one nucleotide with another; may or may not alter amino acid sequences. |
| Deletion mutation | Mutation removing nucleotides, causing frameshift; affects many amino acids downstream. |
| Insertion mutation | Mutation adding nucleotides, causing frameshift; affects many amino acids downstream. |
| Tumour | Mass of abnormal cells resulting from uncontrolled cell division. |
| Cancer | Malignant tumour that invades tissues and can spread (metastasise). |
| Benign tumour | Non-cancerous tumour that does not invade other tissues or metastasise. |
| Malignant tumour | Cancerous tumour capable of invading surrounding tissue and metastasising. |
| Oncogene | Mutated proto-oncogene causing uncontrolled cell division, contributing to cancer development. |
| Tumour suppressor gene | Gene inhibiting cell division and promoting apoptosis; mutations contribute to cancer development when inactivated. |
| Metastasis | Spread of cancer cells from original site to other parts of the body via blood or lymph. |

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| **Crossing over** | **Exchange of genetic material between homologous chromosomes during meiosis, increasing genetic variation.** |
| **Independent assortment** | **Random orientation of homologous chromosome pairs during meiosis I, contributing to genetic variation.** |
| **Acrosome reaction** | **Release of enzymes from sperm acrosome to penetrate egg's zona pellucida during fertilisation.** |
| **Cortical reaction** | **Release of enzymes from cortical granules in egg after fertilisation, preventing polyspermy.** |
| **Zona pellucida** | **Glycoprotein layer surrounding egg cells, involved in preventing polyspermy.** |
| **Haploid cell (n)** | **Cell containing a single set of chromosomes (gametes).** |
| **Diploid cell (2n)** | **Cell containing two sets of chromosomes, one from each parent.** |